

LIBRARY MANAGEMENT SYSTEM

Group 01

Subject: INTE 21343



Team Roles

Product Owners : Surabini, Dinithi

Developers : Pipuni, Bevindu, Udeshika, Pasindu,

QA Engineers : Nadeesha, Aadila

UI/UX Engineers : Baary, Sobashi



System Overview

The Library Management System is a simple application designed to help users and administrators manage library books efficiently

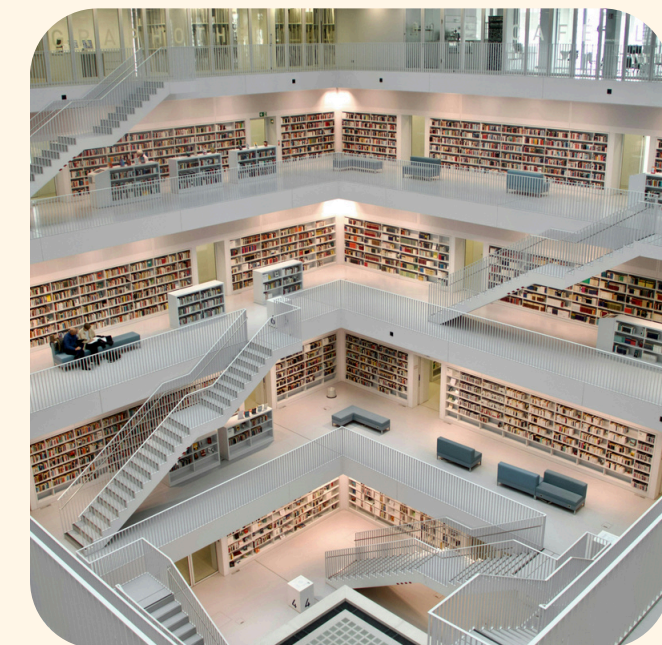
Users can view available books and search by title or author. The system automatically updates book availability in real time after each action.

- Administrators can add new books, remove books, borrow books, and return them and view borrowed books, ensuring the library data stays accurate and up to date.

The system is designed to be user-friendly, fast, reliable, and secure, making it easy to use on basic devices while maintaining correct and consistent data. The development of this system follows the Agile methodology, allowing continuous improvement through iterative sprints

Product Backlog

ID	User Story	Priority	Related Requirement
PB1	As a user, I want to view available books	High	Display book list
PB2	As a user, I want to search books by title or author	High	Search functionality
PB3	As a user, I want to borrow a book	High	Borrow books
PB4	As a user, I want to return a book	High	Return books
PB5	As a system, I want to update book availability in real time	High	Update availability + real-time
PB6	As an admin, I want to add new books	High	Add books
PB7	As an admin, I want to remove books	Medium	Remove books
PB8	As an admin, I want to view borrowed books	Medium	View borrowed books
PB9	As a user, I want a user-friendly interface	Medium	Easy to use UI
PB10	As a system, I want fast response time	Medium	Performance requirement
PB11	As a system, I want to ensure data accuracy	High	Correct book status
PB12	As a system, I want to be reliable without crashes	Medium	Reliability
PB13	As a system, I want to be simple and lightweight	Low	Simplicity
PB14	As a system, I want to run on basic devices	Low	Accessibility
PB15	As a system, I want to be maintainable	Low	Easy updates
PB16	As a system, I want to protect user data	High	Security
PB17	As a system, I want to prevent unauthorized access	High	Authentication/security



SDLC Process



Sprint 1:

Task ID	Task	Assigned Role	Description
S1-T1	Setup project structure	Developer	Create basic HTML/CSS/JS or code files
S1-T2	Create book data	Developer	Add sample book list (array/list)
S1-T3	Display book list	Developer	Show books on screen
S1-T4	Design basic UI	UI/UX	Simple layout (title, list, search bar)
S1-T5	Implement search function	Developer	Search by title/author
S1-T6	Write test cases	QA	Test display & search
S1-T7	Test functionality	QA	Check if search & display works
S1-T8	Define user stories clearly	Product Owner	Finalize Sprint 1 scope

UI Design

Login

Login

User Signup

Sign Up

Sprint 2:

Task ID	Task	Assigned Role	Description
S2-T1	Update book data structure	Developer	Add available status (true/false)
S2-T2	Add Borrow button	Developer	Add button for each book
S2-T3	Implement Borrow function	Developer	Change status to unavailable
S2-T4	Add Return button	Developer	Allow returning books
S2-T5	Implement Return function	Developer	Change status to available
S2-T6	Update UI for status	UI/UX	Show Available / Borrowed
S2-T7	Improve UI layout	UI/UX	Make buttons clear and user-friendly
S2-T8	Create test cases	QA	<u>Test</u> borrow/return actions
S2-T9	Test functionality	QA	Check correct status updates
S2-T10	Validate features	Product Owner	Confirm requirements met

UI Design

Library System (User)

Harry Potter
Author: J.K. Rowling
Available: 3 / 5

The Hobbit
Author: J.R.R. Tolkien
Available: 4 / 4

1984
Author: George Orwell
Available: 0 / 2

Sprint 3:

Task ID	Task	Assigned Role	Description
S3-T1	Create admin login (basic)	Developer	Simple username/password check
S3-T2	Add "Add Book" feature	Developer	Admin can add new books
S3-T3	Add "Remove Book" feature	Developer	Admin can delete books
S3-T4	Display borrowed books	Developer	Show list of borrowed books
S3-T5	Improve UI for admin panel	UI/UX	Simple admin section design
S3-T6	Add validation	Developer	Prevent empty inputs & errors
S3-T7	Security check	Developer	Prevent unauthorized access (basic login)
S3-T8	Create final test cases	QA	Test all features (full system)
S3-T9	System testing	QA	Check full system works correctly
S3-T10	Final validation	Product Owner	Approve final product
S3-T11	Add "Sign Up" feature	Developer	Allow new users to create account

UI Design

Library System (Admin)

Search by title or author...

Add Book

Book Title

Author

Total Copies

Add Book

Controls

Show All Books

Show Borrowed Books

Harry Potter

Author: J.K. Rowling

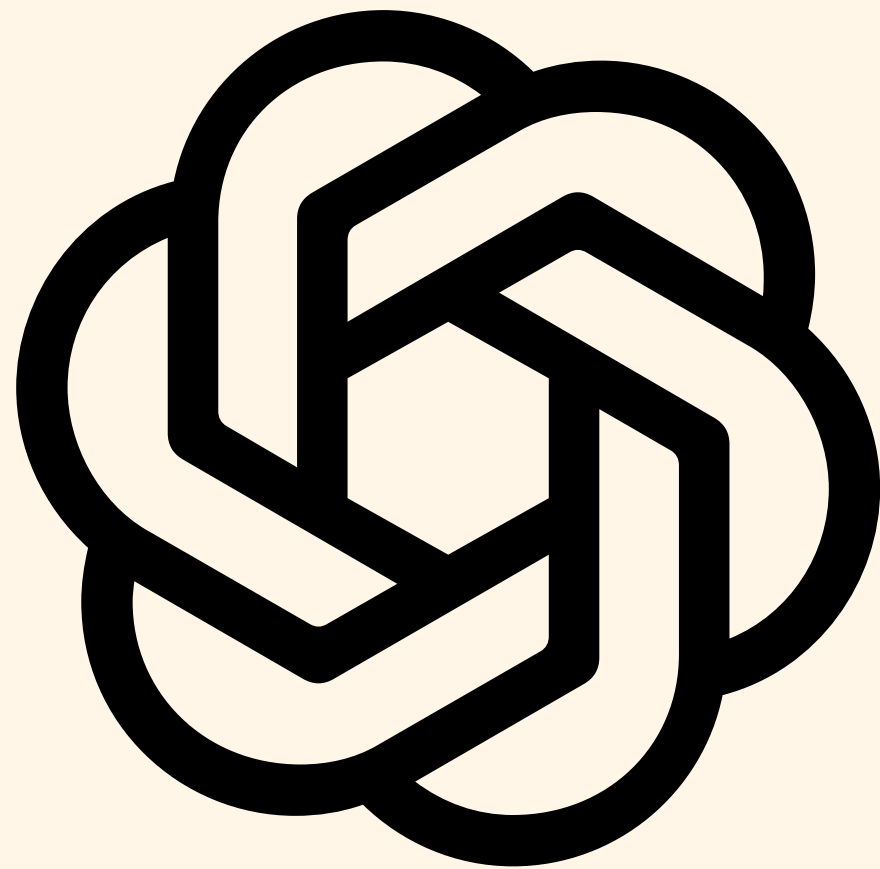
Available: 3 / 5

Return

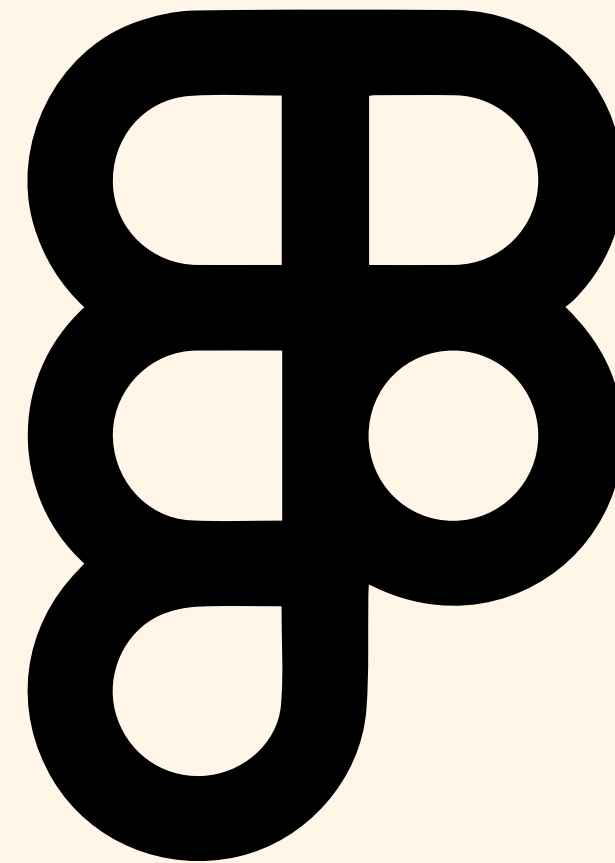
Delete

Borrow

LLM for Development



ChatGPT



Figma

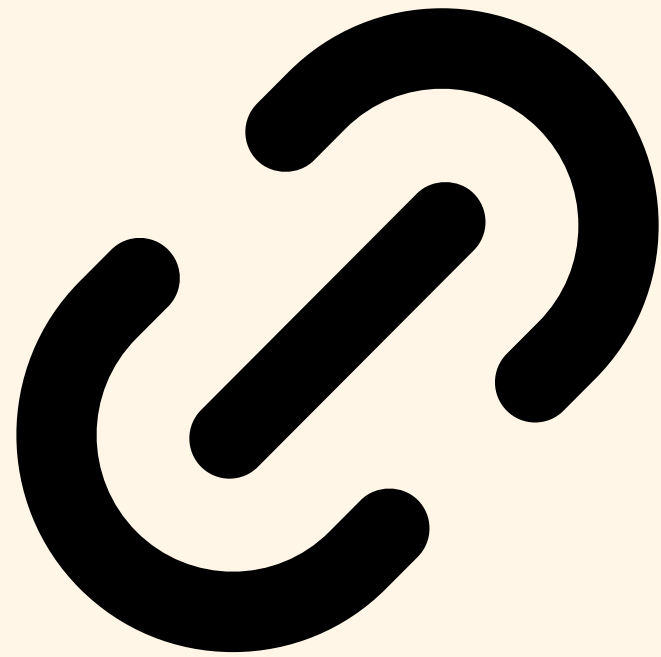
Challenges

- Time limitation (only 3 hours)
- Understanding requirements clearly
- Integrating UI with functionality
- Using AI code correctly (sometimes confusing)
- Handling real-time updates
- Keeping the system simple but functional

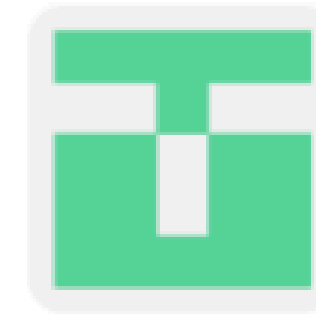
✧ ✧ The Learnings ✧ ✧

- Importance of teamwork and communication
- Importance of planning before coding
- Breaking system into small tasks (sprints)
- Understanding requirements clearly before development
- Importance of simple and clean design
- How to use AI tools effectively and responsibly
- Importance of testing early (not at the end)
- Time management is critical
- Collaboration between roles (Dev, QA, UI/UX, PO)

GitHub Repository



BIFernando/SEC- ASSIGNMENT



Role-Based SDLC Simulation Project

1
Contributor

0
Issues

0
Stars

0
Forks



BIFernando/SEC-ASSIGNMENT: Role-Based SDLC Simulation Project

Role-Based SDLC Simulation Project. Contribute to BIFernando/SEC-ASSIGNMENT development by creating an account on GitHub.

 GitHub



THANK YOU!

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IM/2023/003
IM/2023/069
IM/2023/083
IM/2023/062
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